# Response surface alignment measure

## 1 Set-up

AOD and ARI at the gridbox at longitude -171.5625 and latitude 51.875 (shown in green in Figure 1) are considered, for July.

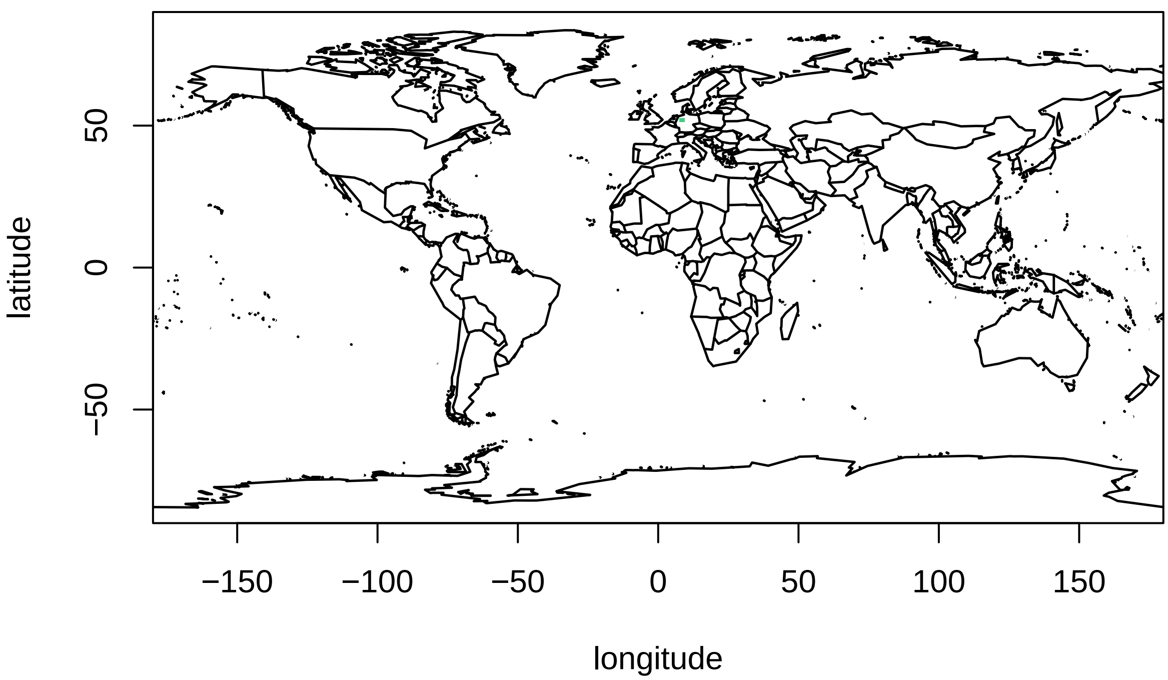


Figure 1: Selected gridboxes highlighted in green and blue.

# 2 Sensitivity analysis

GAM-based sensitivity analysis is performed for the two outputs. To reach 120% of the variance of the two outputs requires 37 and 31 inputs respectively (with main effects summing to 115% and 120%); see Table 1 and Figure 2..

Table 1: Post-SA main effects for the first (left) and second (right) outputs.

| Input | Main effect (%) | Input | Main effect (%) |
|---|---|---|---|
| dry_dep_acc | 42 | prim_so4_diam | 29 |
| prim_so4_diam | 20 | dry_dep_so2 | 22 |
| bvoc_soa | 13 | dry_dep_acc | 12 |
| kappa_oc | 9 | dry_dep_ait | 7 |
| sea_spray | 6 | anth_so2_eur | 7 |
| dry_dep_so2 | 4 | anth_so2_r | 6 |
| anth_so2_eur | 3 | bc_ri | 5 |
| bparam | 3 | anth_so2_chi | 5 |
| dbsdtbs_turb_0 | 2 | anth_so2_asi | 4 |
| anth_so2_chi | 2 | kappa_oc | 3 |
| c_r_correl | 2 | m_ci | 3 |
| bc_ri | 2 | cloud_ph | 2 |
| oxidants_oh | 1 | dbsdtbs_turb_0 | 2 |
| dry_dep_ait | 1 | anth_so2_nam | 2 |
| ai | 1 | sea_spray | 2 |
| carb_ff_diam | 1 | bparam | 2 |
| prim_moc | 0 | a_ent_1_rp | 1 |
| ait_width | 0 | oxidants_oh | 1 |
| autoconv_exp_lwp | 0 | carb_res_diam | 1 |
| anth_so2_asi | 0 | c_r_correl | 1 |
| conv_plume_scav | 0 | two_d_fsd_factor | 1 |
| volc_so2 | 0 | conv_plume_scav | 1 |
| dms | 0 | autoconv_exp_nd | 0 |
| a_ent_1_rp | 0 | ai | 0 |
| oxidants_o3 | 0 | carb_ff_diam | 0 |
| two_d_fsd_factor | 0 | oxidants_o3 | 0 |
| m_ci | 0 | bl_nuc | 0 |
| carb_bb_diam | 0 | prim_moc | 0 |
| cloud_ph | 0 | sig_w | 0 |
| anth_so2_nam | 0 | volc_so2 | 0 |
| cloud_ice_thresh | 0 | rain_frac | 0 |
| bl_nuc | 0 | | |
| rain_frac | 0 | | |
| autoconv_exp_nd | 0 | | |
| sig_w | 0 | | |
| anth_so2_r | 0 | | |
| carb_res_diam | 0 | | |

An emulator will be fitted for the union of inputs from Table 1, namely the 37 inputs:

1. dry_dep_acc
2. prim_so4_diam

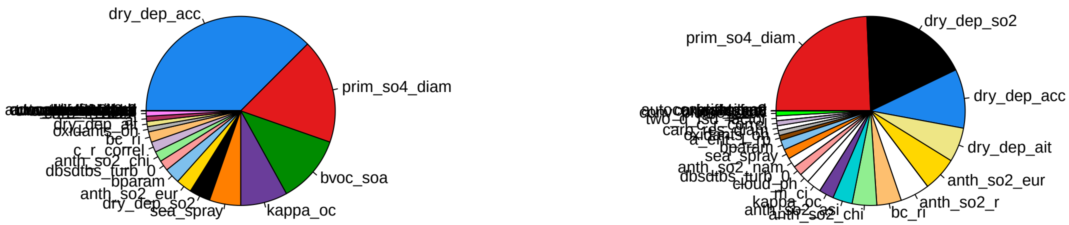


Figure 2: The variables with main effects summing to at leaat 95% for (left) output 1 and (right) output 2.

3. bvoc_soa
4. kappa_oc
5. sea_spray
6. dry_dep_so2
7. anth_so2_eur
8. bparam
9. dbsdtbs_turb_0
10. anth_so2_chi
11. c_r_correl
12. bc_ri
13. oxidants_oh
14. dry_dep_ait
15. ai
16. carb_ff_diam
17. prim_moc
18. ait_width
19. autoconv_exp_lwp
20. anth_so2_asi
21. conv_plume_scav
22. volc_so2
23. dms
24. a_ent_1_rp
25. oxidants_o3
26. two_d_fsd_factor
27. m_ci
28. carb_bb_diam
29. cloud_ph
30. anth_so2_nam
31. cloud_ice_thresh
32. bl_nuc
33. rain_frac
34. autoconv_exp_nd
35. sig_w
36. anth_so2_r
37. carb_res_diam

# 3 Emulator fit and validation

To fit the emulator, we assume a linear mean function of the form

$$m(\mathbf{x}) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p.$$

The Gauss (squared exponential) kernel is used for the covariance function, maximum posterior mode estimation for the parameters estimates (trend, range, SD and the nugget term), and the `RobustGaSP` package to fit the emulators.

For validation of the emulators fitted, as well as calculation of the alignment measure, 100 new LHC points are generated and predictions made at these points. The mean width of the 95% CI's for these predictions are shown in Table 3, along with the mean width of the LOO CV prediction 95% CI's. Note that the range of output 1 is 0.6 and of output 2 is 2.6.

Table 2: Validation checks

| Output | Prediction 95% CI mean width | LOO CV 95% CI mean width | % validated in LOO CV |
|---|---|---|---|
| 1 | 0.23 | 0.23 | 95 |
| 2 | 0.92 | 0.92 | 96 |

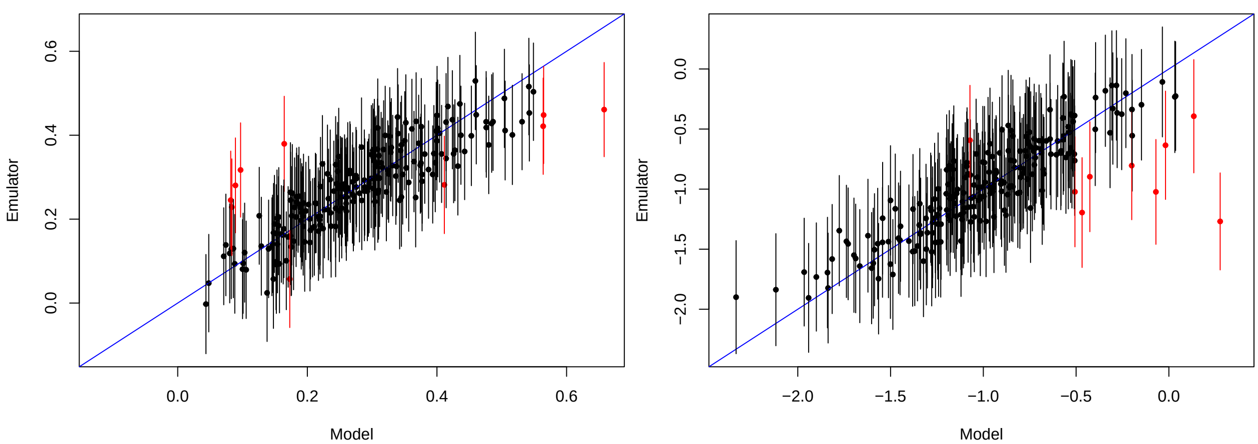


Figure 3: LOO CV validation plot for output 1 (left) and output 2 (right).

# 4 Check of by-hand calculations

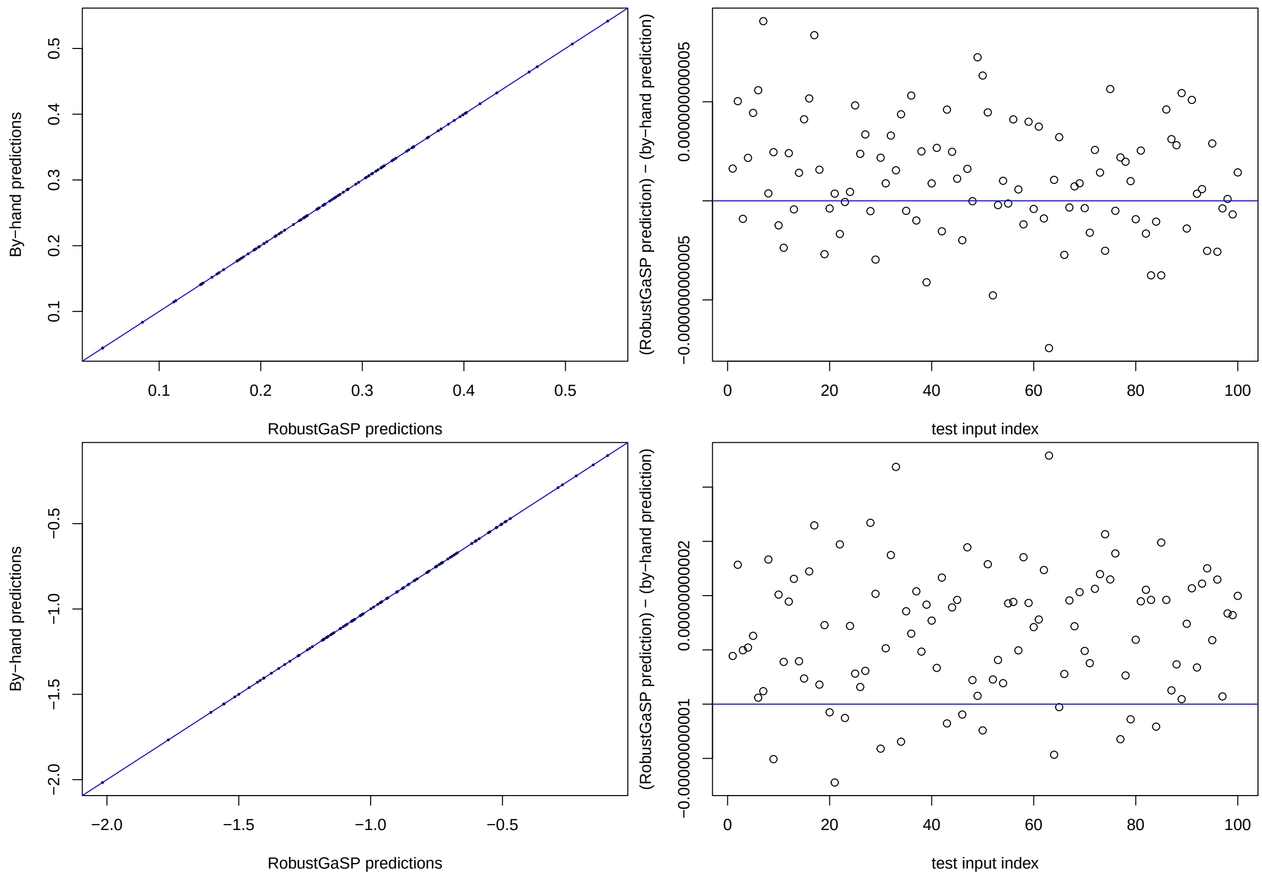


Figure 4: The predictions found by-hand match well with those made by the RobustGaSP package (left), with the latter minus the former found to be essentially zero (right). Top: Output 1. Bottom: Output 2.

Ahead of performing by-hand calculations of the partial derivatives at the testing points, by-hand calculations for the predictions made at these points are plotted against the predictions returned by the `RobustGaSP` package (left hand plots in Figure 3) and the difference between the two plotted against the test input index (right hand plots in Figure 3). Finally, for both we calculate

$$\frac{\sum_{i=1}^{N}\left|(\text{predictions from package})_i - (\text{predictions by-hand})_i\right|}{N},$$

where $N$ is the number of test points to get 0 for output 1 and 0 for output 2.

# 5 Calculation of the alignment measure

Having calculated the emulated partial derivatives for use in the alignment measure, and normalising these, we calculate the AM, getting a value of 0.535.